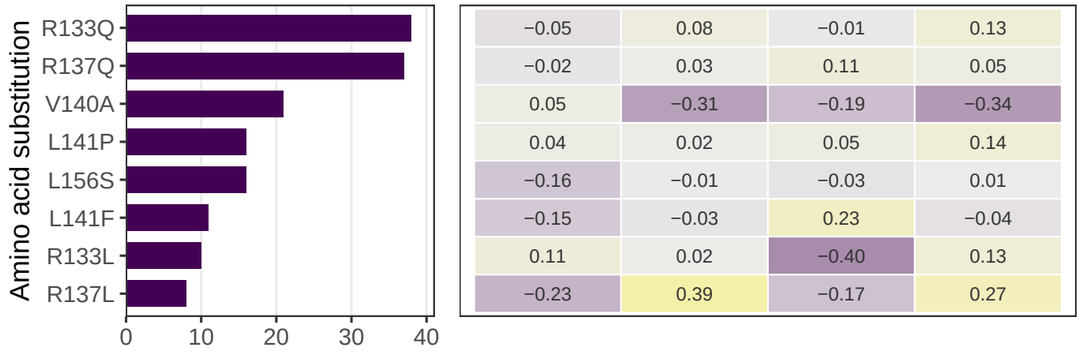
**A**

*H1N1 − Ridge*

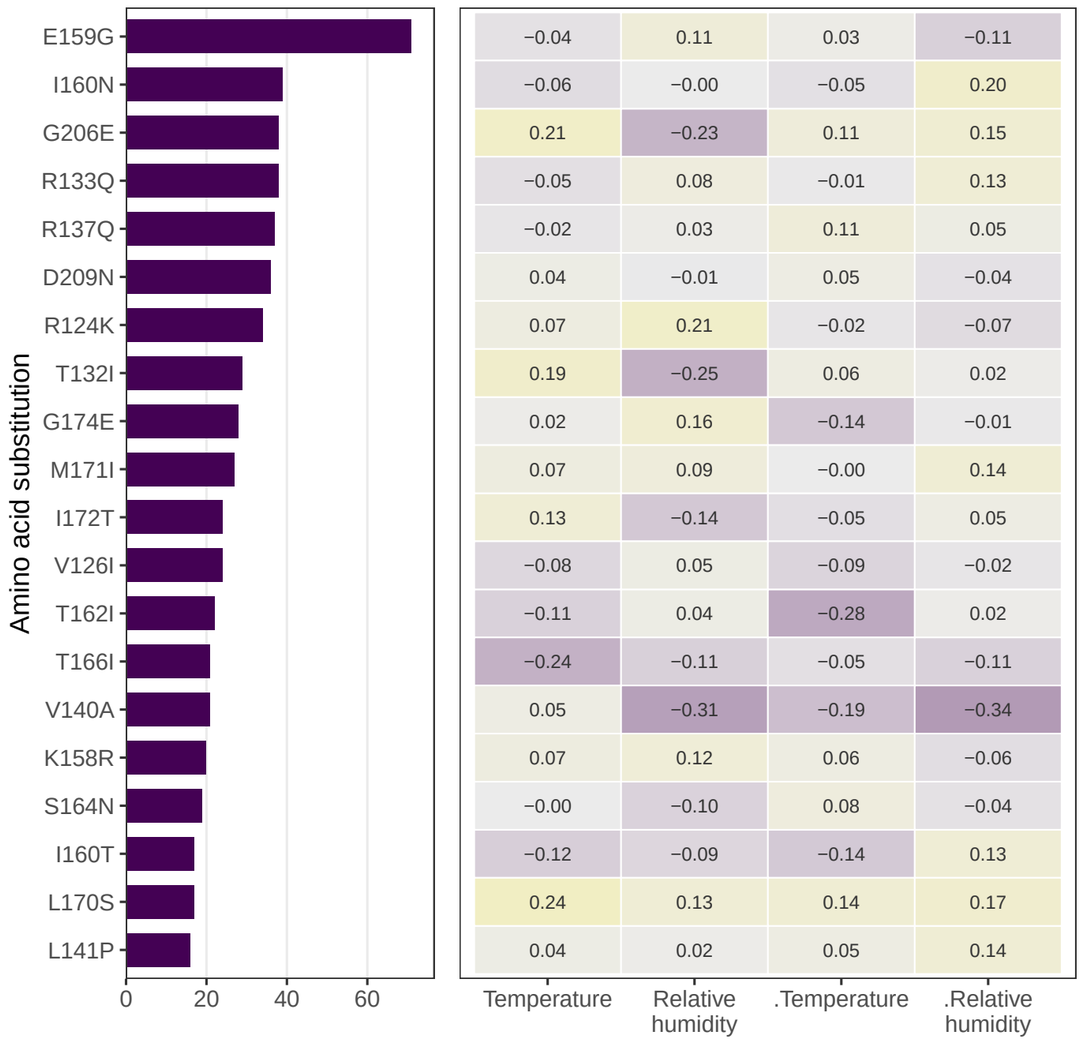



**B**

*H3N2 − Ridge*

**C**

*H1N1 − Epitope (top 20 mutations)*




**D**

*H3N2 − Epitope (top 20 mutations)*